

Forensic Report - Fingerprint Comparison

Fingerprint Analysis Workstation - Version 2.3

Report generated at: 2026-05-06 21:34:29

Lab disclaimer: This report contains statistical software outputs and must not be used as sole identity evidence.

Audit Trail

Case Reference: -

Operator: -

SHA-256 (Reference file):
cd22f87c92a9af937ae7f7eb8dee83ebbb48bd029c22ad06265c6a6a498a8645

SHA-256 (Query file):
a274e2a7fb14496aa766228624c149392fd0225d072b9be07d4cf2869fc2b4eb

Methodology Summary

The analysis pipeline consists of several key steps: 1. Image Preprocessing: The input images are first processed using CLAHE (Contrast Limited Adaptive Histogram Equalization) to enhance contrast. This is followed by a thinning operation to extract the skeleton of the ridges. 2. Feature Extraction: The processed images are then analyzed using a combination of Otsu's method for thresholding and Gabor filters for ridge orientation and frequency analysis. 3. Comparison: The extracted features are compared using a statistical method to determine the likelihood of the two fingerprints being from the same source. The final result is a statistical output indicating the degree of similarity.

Recorded Run Parameters

Parameter	Value
border_margin	12
min_distance	14
min_contrast	20

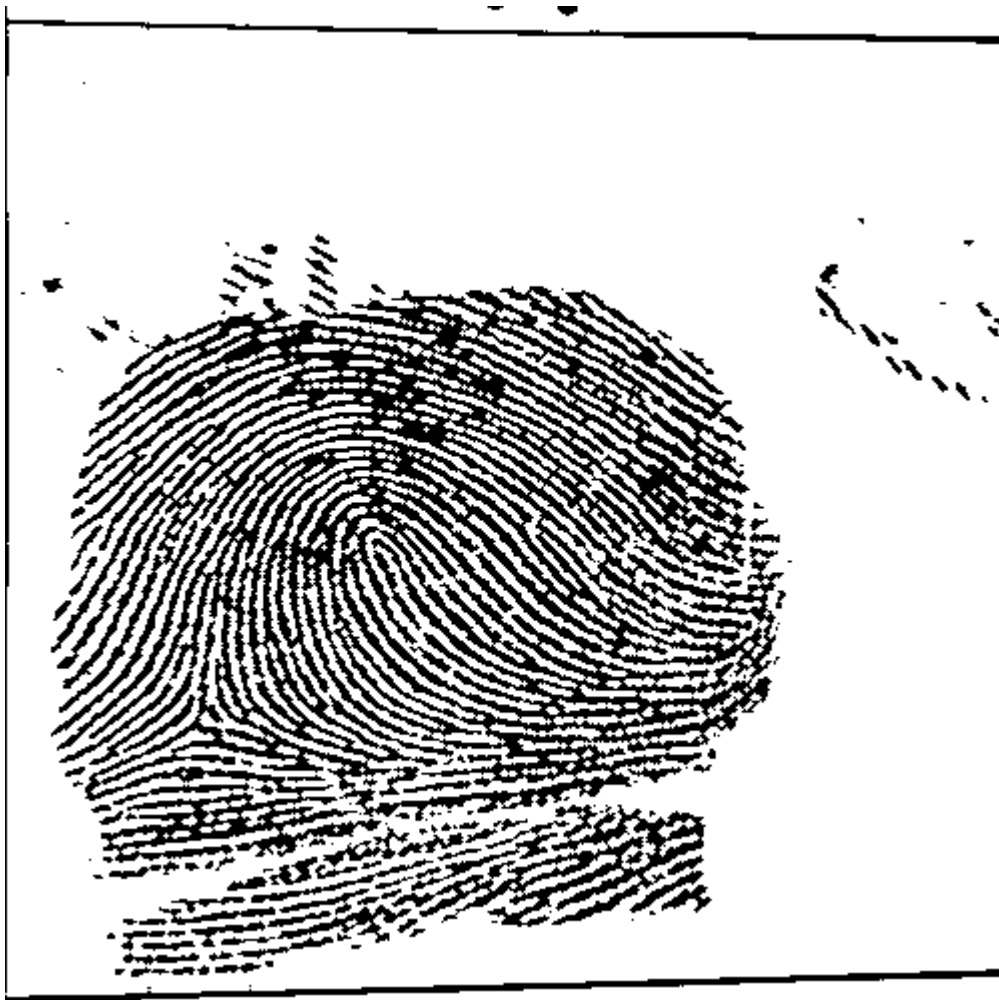
min_angle_diff	12
denoise_method	fastNlMeans
fast_denoise_h	10
gauss_ksize	3
MATCH_DISTANCE_THRESHOLD	15
MATCH_ANGLE_THRESHOLD_DEG	35
PARTIAL_VERIFY_SEARCH_RADIUS	80
PARTIAL_VERIFY_STEP_PX	10

Pipeline Images

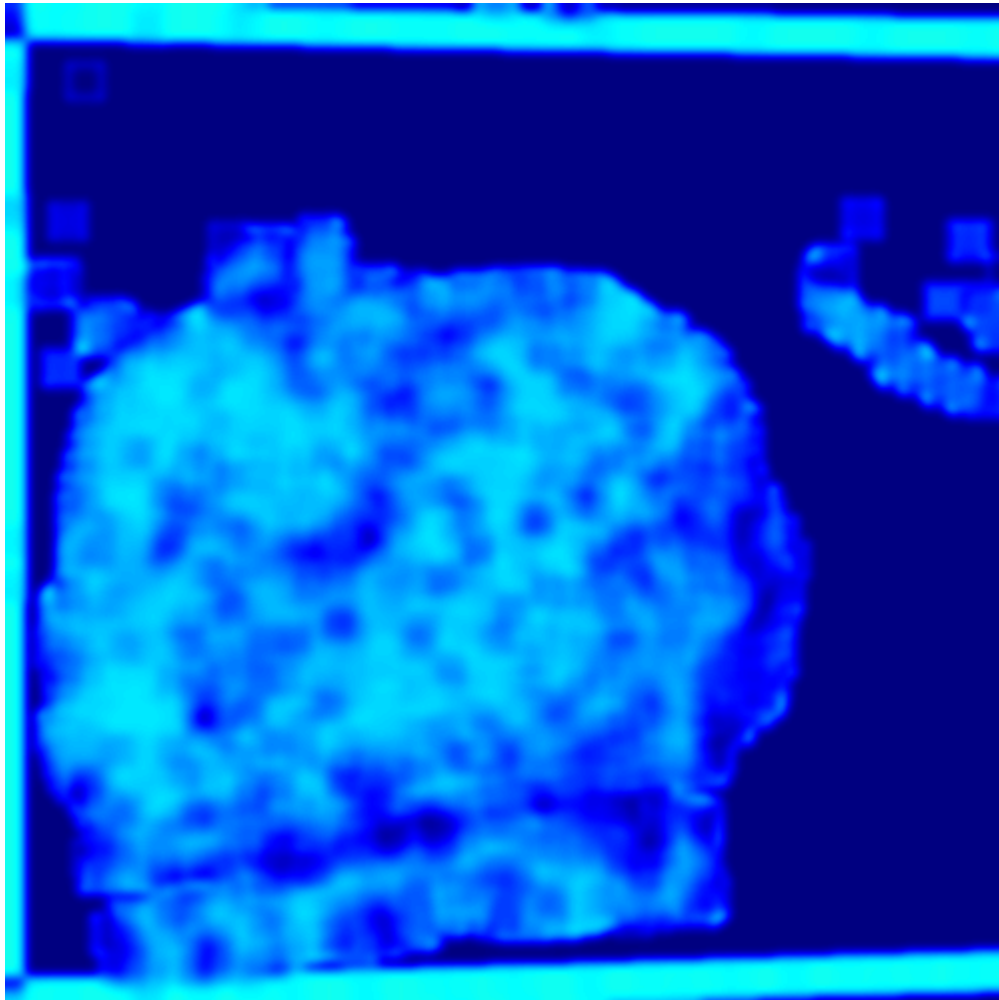
Reference Processing Pipeline

Stats: Binary white pixels: 195963 | Gabor white pixels: 12787 | Skeleton white pixels: 6802 | Filtered minutiae: 154

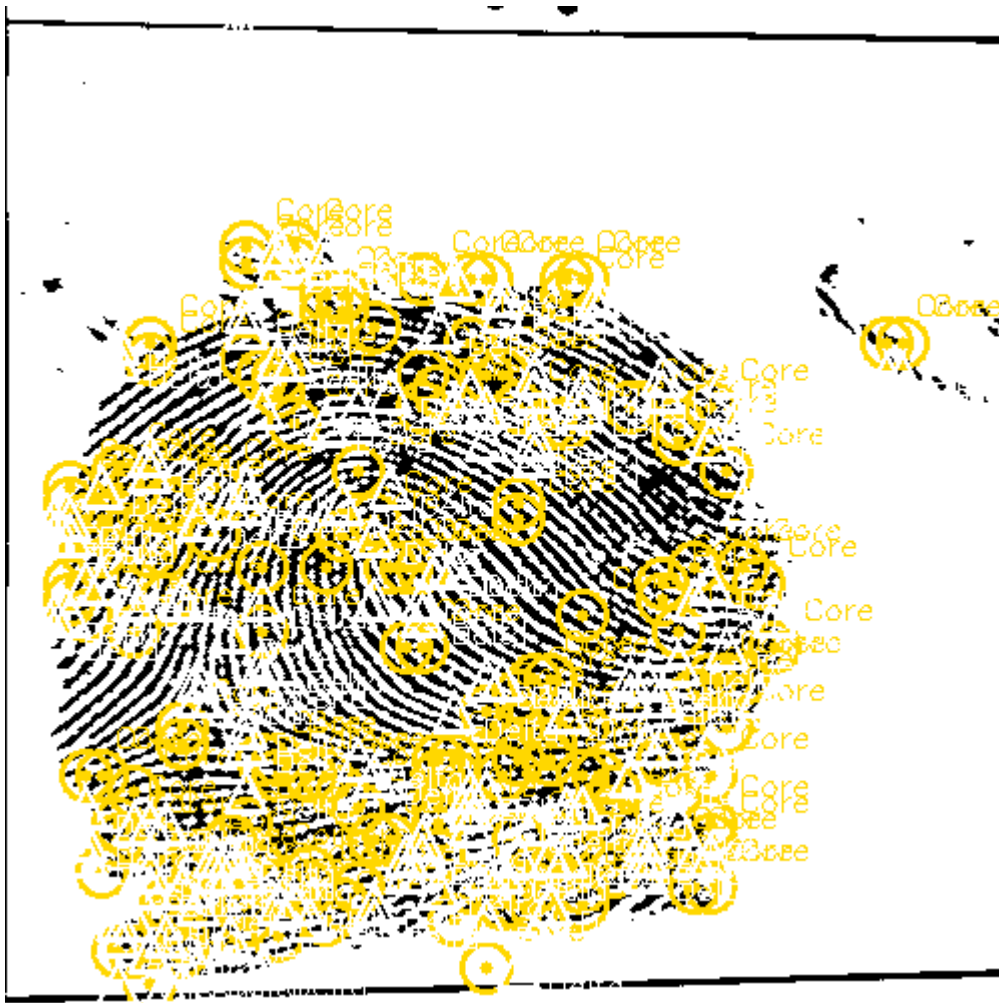
Preprocessed Binary



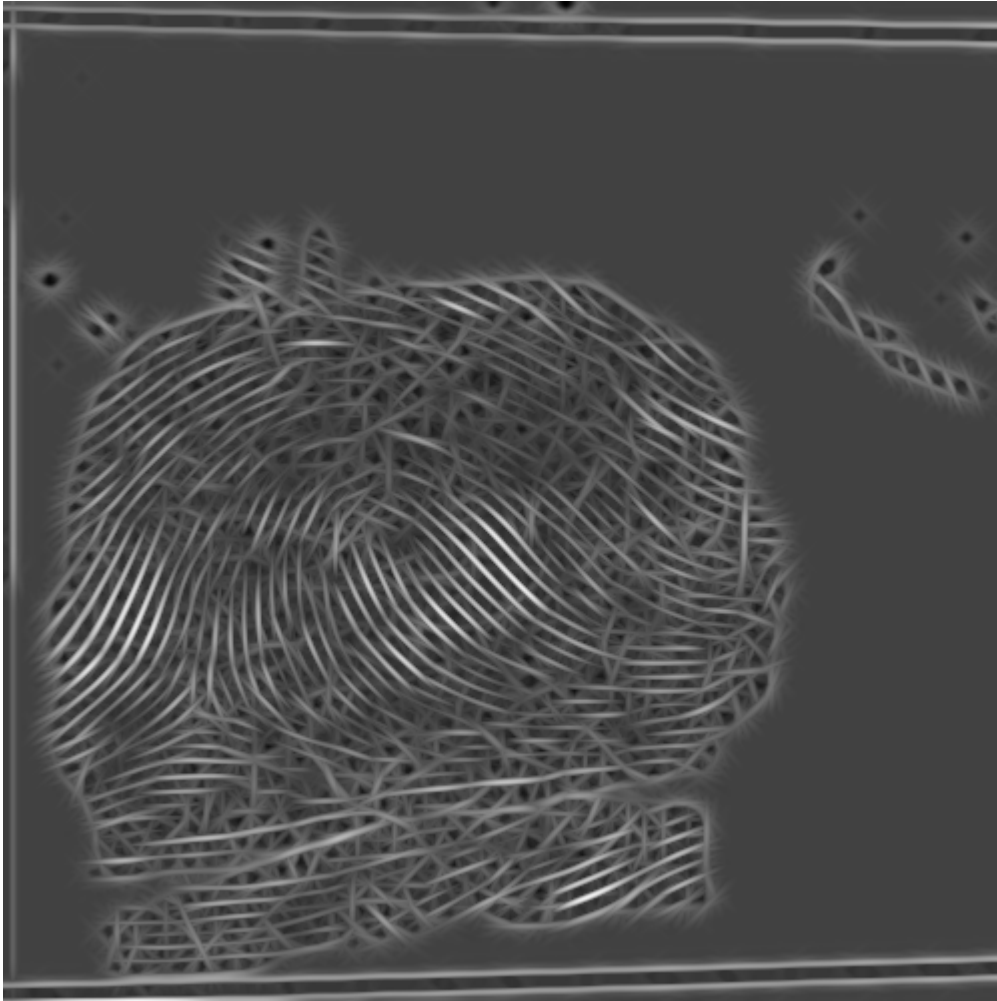
Quality Heatmap



Singular Points



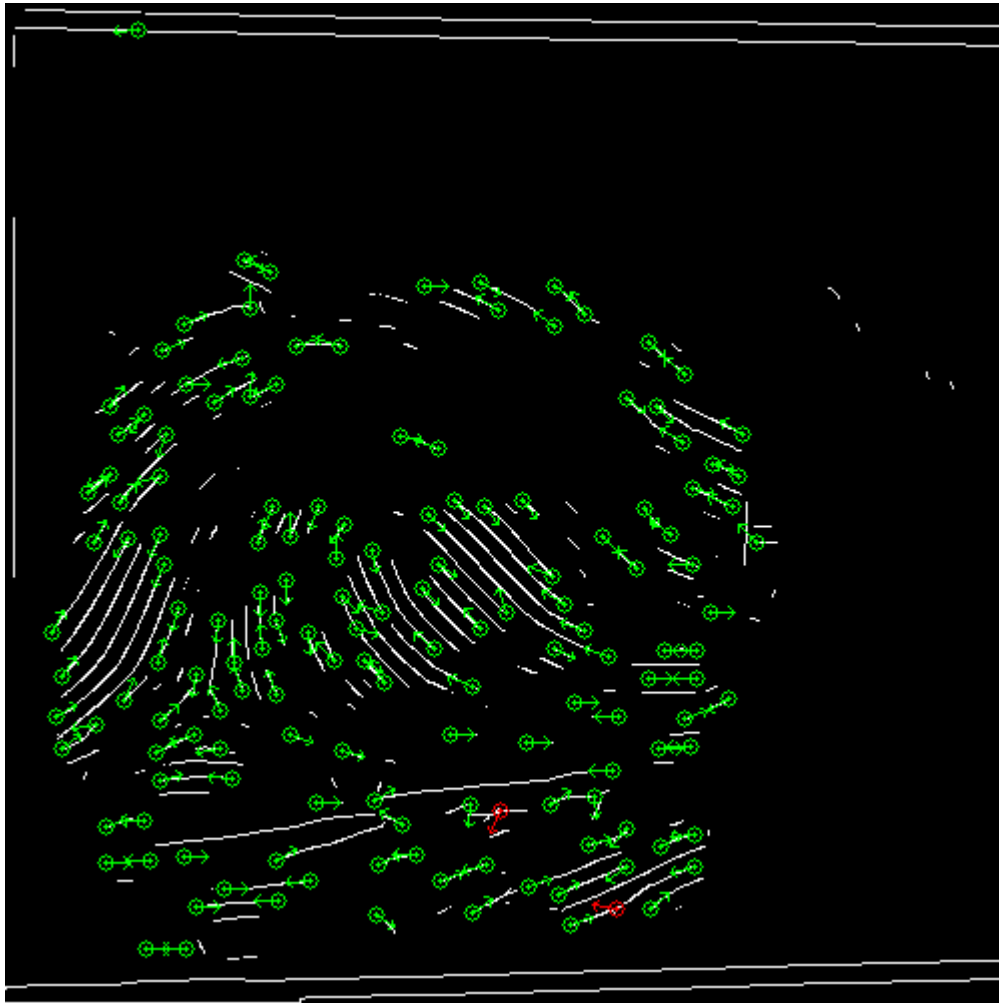
Ridge Enhancement (Gabor)



Skeleton



Minutiae Visualization



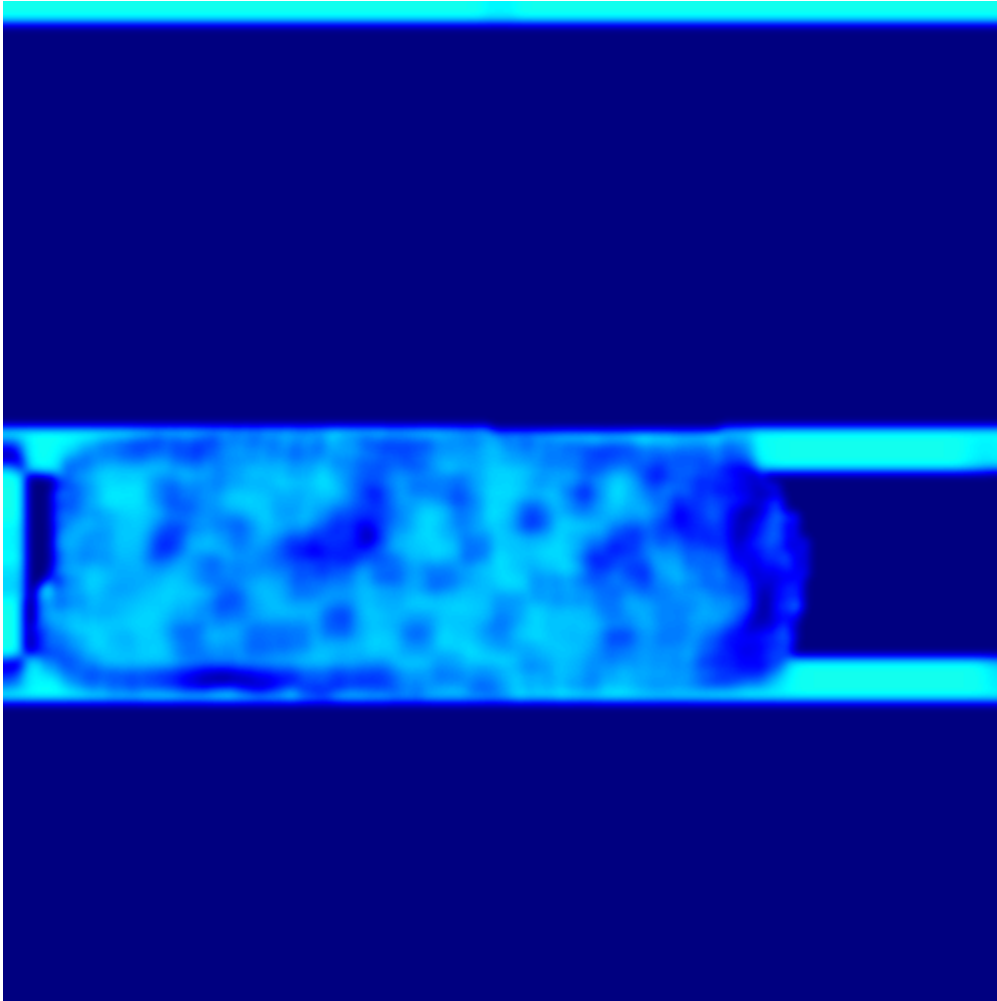
Query Processing Pipeline

Stats: Binary white pixels: 225356 | Gabor white pixels: 7571 | Skeleton white pixels: 3720 | Filtered minutiae: 67

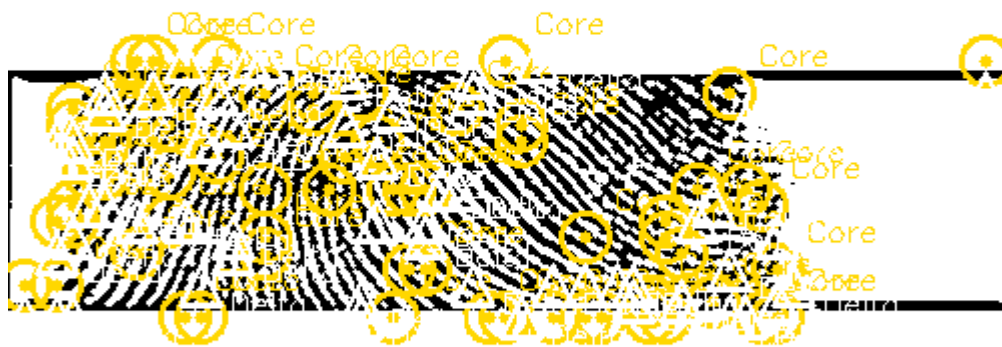
Preprocessed Binary



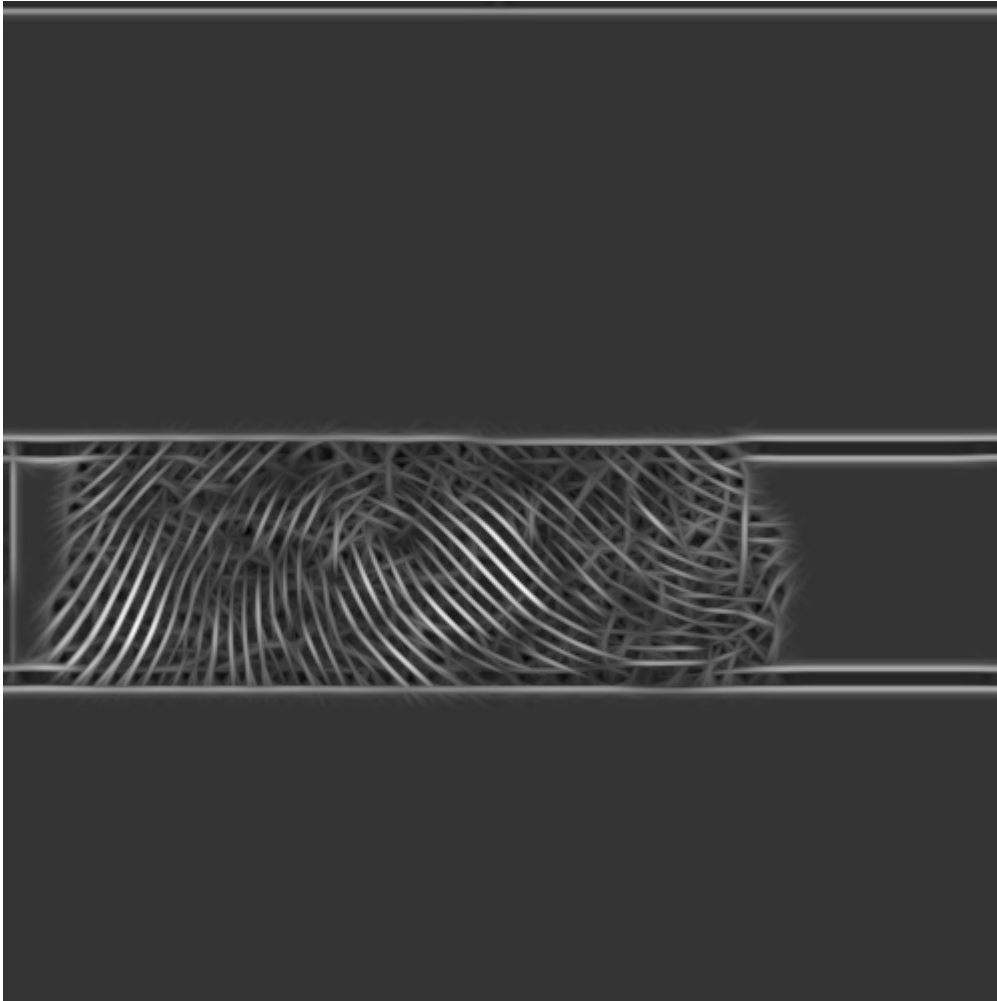
Quality Heatmap



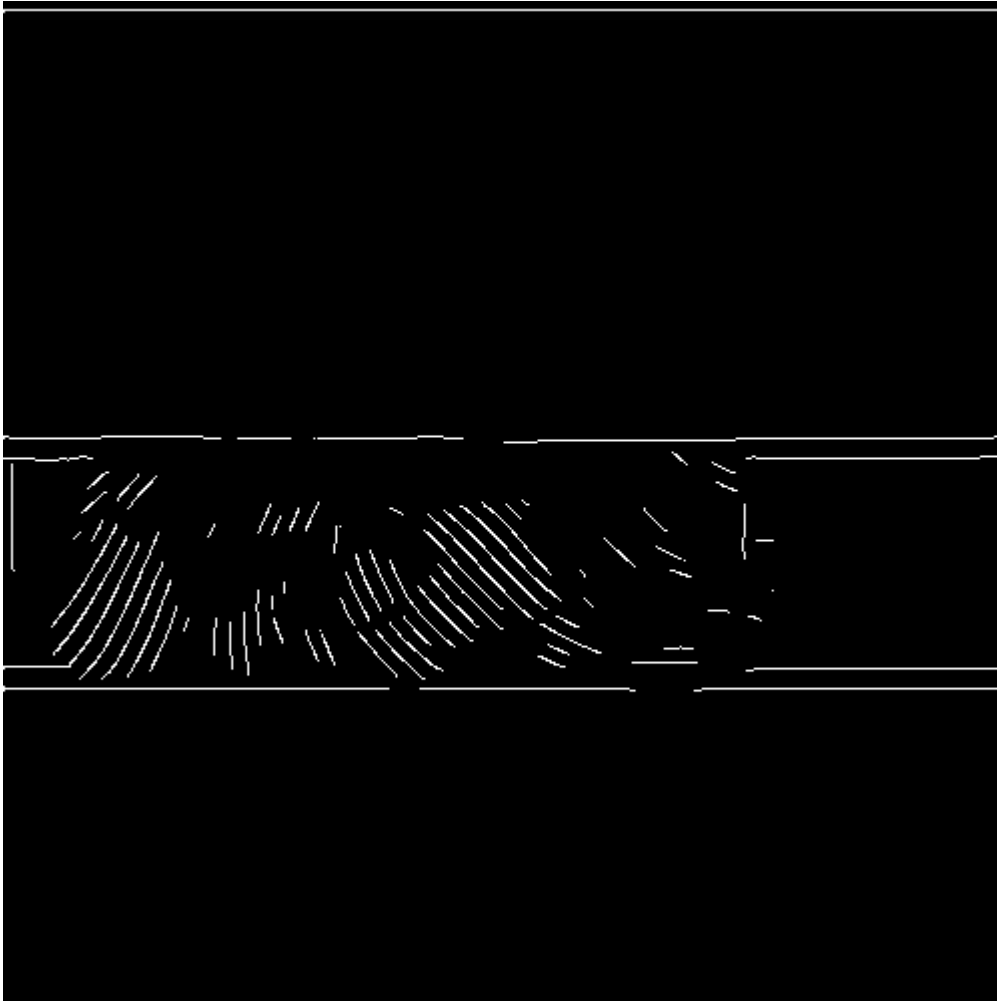
Singular Points



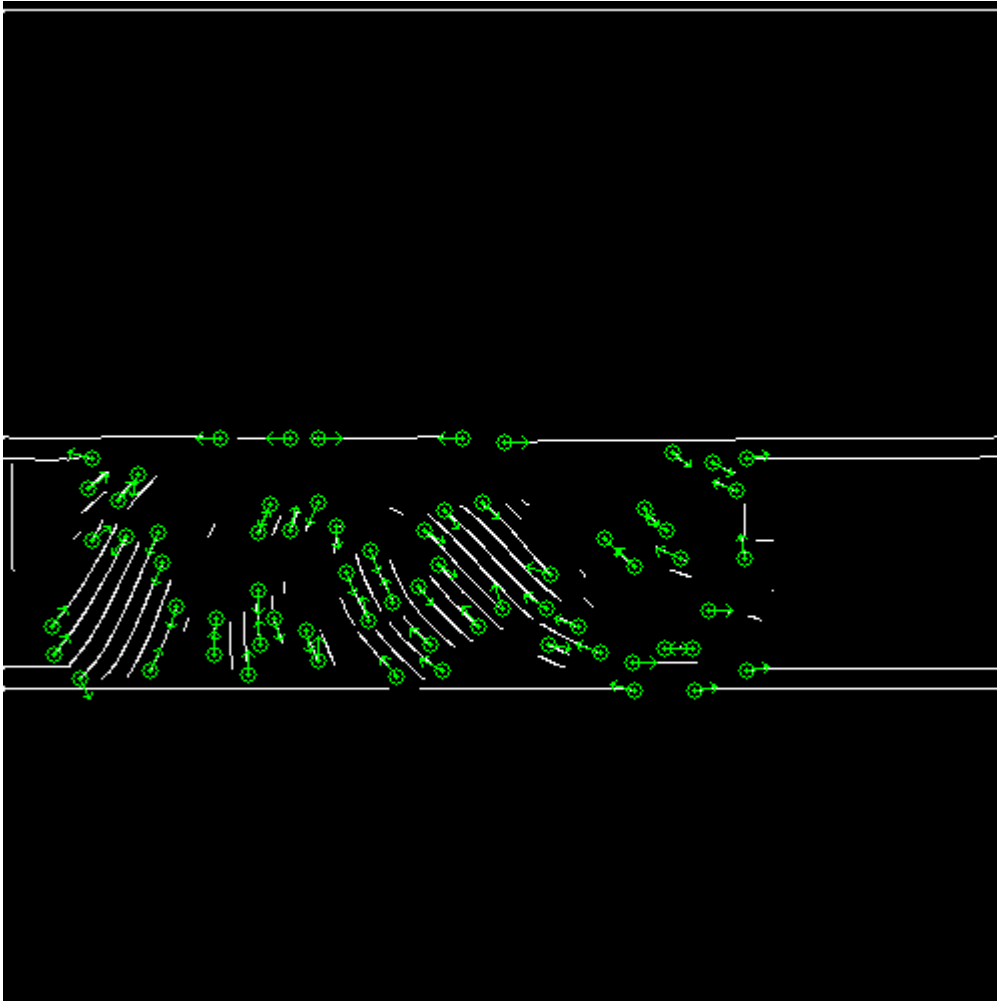
Ridge Enhancement (Gabor)



Skeleton

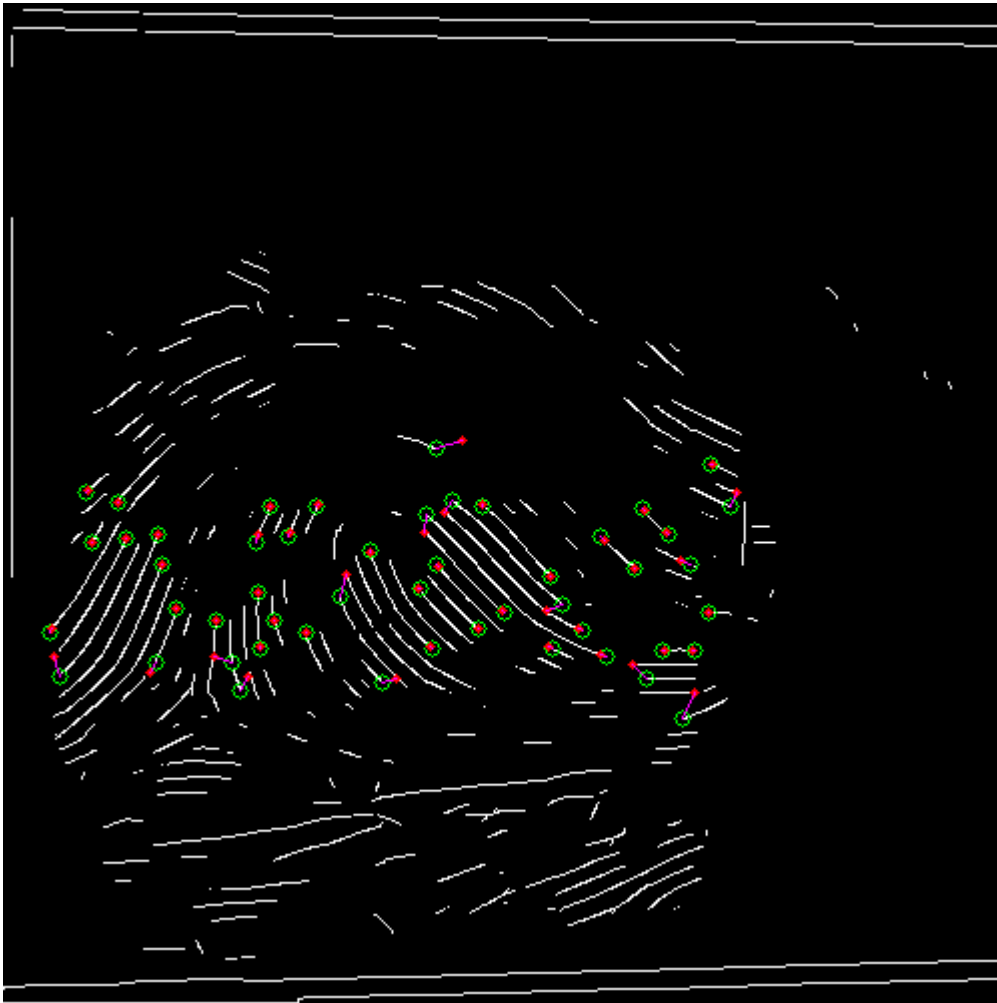


Minutiae Visualization



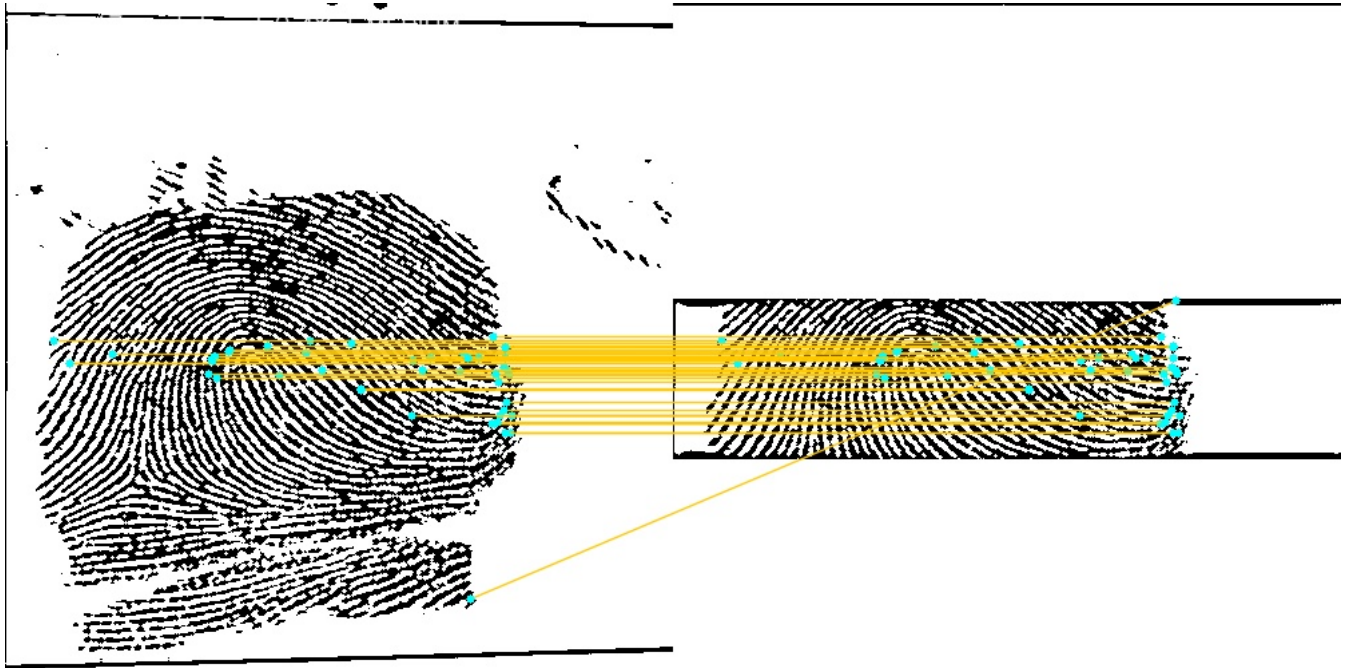
Aligned Match Visualization

Match Overlay



ORB Visual Verification

ORB Feature Matches

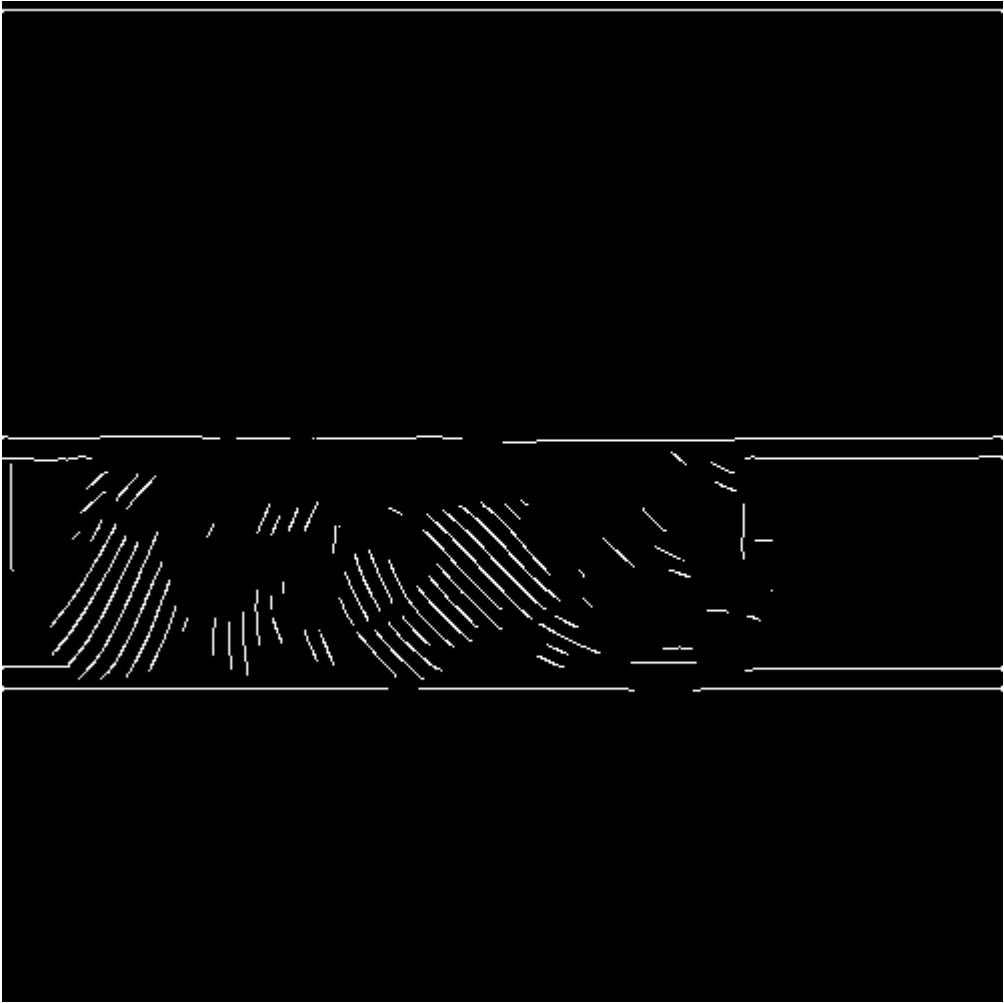


Final Skeletons Used for Matching

Reference Skeleton



Query Skeleton



Archive PNG files: original_skeleton.png, partial_skeleton.png in report folder.

Partial-to-Reference Alignment

■■■■■■ ■■■■■■ ■■■■ ■■■■■■■■■■ (■■■■■■■■■■): ■■■■■■ dx=0 dy=0 ■■■■■■ 0.0°. ■■■■■■■■■■ ■■■■ ■■■■■■■■■■: 50 — ■■■■
■■■■■ ■■■■■■■■: 50 (■■■■■■■ 0) ■■■■■■ ■■■■■■ ■■■■■■■■■■: +0.00 ■■■■■■.

Green: matched reference points | Orange: matched partial points after alignment.

Statistical Comparison Results

Reference minutiae count: 154

Query minutiae count: 67

One-to-one matched pairs: 50

Match score (query-based): **74.63%**

Fused Score: **56.43%**

Fusion Components - Minutiae: **74.63%** | MCC: **35.72%** | ORB: **9.40%**

[illegible]

Dice score: **45.25%**

Forensic tier: ■■■■■ ■ — ■■■■ ■■■■■ ■■■■■■■ ■■■■■■■ ■■■ ■■■■■ ■■■■■■■ ■■■■■■■■■

[illegible]

Internal status: HIGH MATCH